

Chapter 2: Built-in Data Structures Under the Hood

Core Learning Objective: Peel back the syntax of Python lists, dictionaries, tuples, sets, and deques to understand their underlying C structures, memory layout, dynamic resizing algorithms, and amortized Big-O complexities.

1. Python Lists: Contiguous Pointer Arrays

Zero-Prerequisite Intuition: The "Coat-Check Ticket Rack" Metaphor What is a Python `list` under the hood?

Imagine you run the coat-check room at a luxury theatre:

- Guest 1 checks in a giant heavy winter fur coat.
- Guest 2 checks in a tiny pair of sunglasses.
- Guest 3 checks in an umbrella.

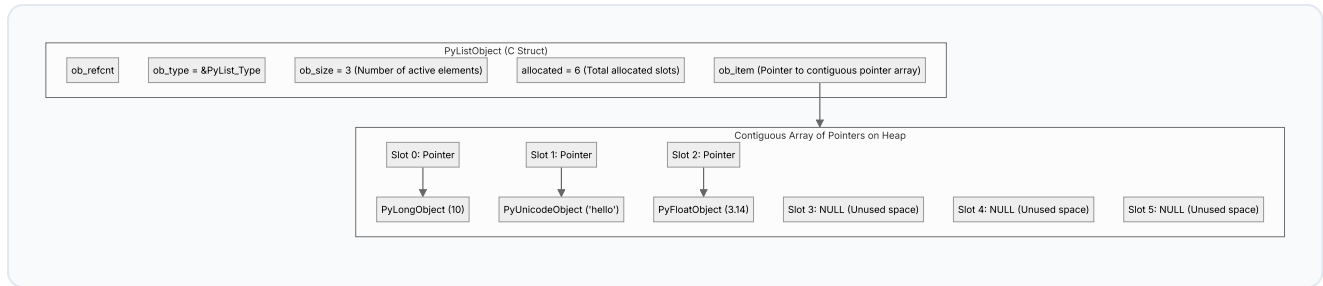
You cannot glue the fur coat, the sunglasses, and the umbrella together in a straight line—they are completely different sizes and shapes!

What do you do instead? You have a neat, orderly rack of **identical wooden hangers numbered 0, 1, 2, 3...**

- On Hanger #0, you hang a claim ticket pointing to the winter fur coat in the back room.
- On Hanger #1, you hang a claim ticket pointing to the sunglasses.
- On Hanger #2, you hang a claim ticket pointing to the umbrella.

In Python, a `list` does **NOT** hold the actual data items! A Python `list` is simply an **orderly rack of identical claim tickets (memory pointers)**. Each hanger is exactly 8 bytes wide, and each hanger points to wherever that object is floating on the warehouse floor (the Heap)!

A Python `list` is **not a linked list**. It is a **dynamically sized, contiguous array of pointers** (`PyObject**`).



The CPython Definition (`listobject.h`)

Dynamic Resizing & Over-Allocation Formula

When you call `list.append()`, CPython checks if `ob_size == allocated`. If the buffer is full, it resizes the array using this exact formula:

$$\text{new_allocated} = \text{new_size} + (\text{new_size} \gg 3) + (\text{new_size} < 9 ? 3 : 6)$$

Key Takeaways:

- As the list grows, capacity increases in bursts: $0 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 24 \rightarrow 32 \dots$
- Because resizing occurs geometrically, appending has an **amortized time complexity of $O(1)$** , even though an individual resize operation costs $O(N)$ memory copies.

Time Complexity Matrix: List Operations

Operation	Average Case	Worst Case	Memory / Mechanism
<code>list.append(x)</code>	$O(1)$	$O(N)$ (rare resize)	Appends to end of array
<code>list.pop()</code>	$O(1)$	$O(1)$	Decrements <code>ob_size</code>
<code>list.insert(0, x)</code>	$O(N)$	$O(N)$	Shifts all N pointers right by 1
<code>list.pop(0)</code>	$O(N)$	$O(N)$	Shifts all $N - 1$ pointers left by 1
<code>list[i]</code> (Lookup)	$O(1)$	$O(1)$	Direct pointer arithmetic: <code>ob_item + i * sizeof(void*)</code>
<code>x in list</code> (Search)	$O(N)$	$O(N)$	Linear iteration over pointers

2. Python Tuples: Fixed Memory Footprint

A `tuple` is an **immutable contiguous pointer array**. Because its length is fixed at instantiation, CPython eliminates the `allocated` capacity field.

The Tuple Free List Optimization

To accelerate code execution, CPython maintains an internal **free list** for small tuples (up to 20 elements). When a tuple is destroyed, its memory is not immediately freed back to the OS; instead, it is retained in a freelist to instantly satisfy future tuple allocations without calling `malloc()`.

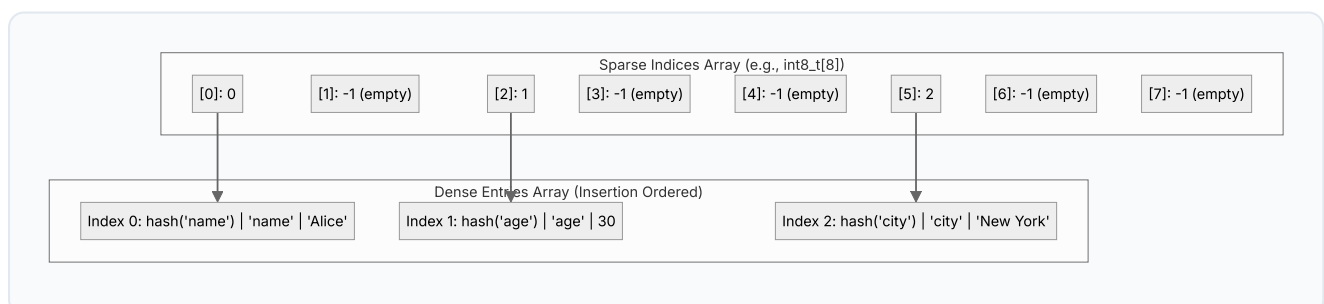
3. The Modern Dictionary: Compact Hash Tables

Prior to Python 3.6, dictionaries used a sparse hash table where each entry took 24 bytes (hash, key pointer, value pointer), resulting in vast amounts of wasted unused memory:

The Raymond Hettinger Compact Dict (Python 3.6+)

Modern Python separates the sparse table into **two distinct arrays**:

1. **Indices Array (Sparse)**: An array of tiny integers (1 byte for small dicts: `int8_t`).
2. **Entries Array (Dense)**: A compact array storing only active entries in order of insertion.



Why Dicts Preserve Insertion Order

Because new keys are simply appended to the end of the **Dense Entries Array**, iterating over a dictionary merely walks the entries array sequentially. This is why Python dictionaries are guaranteed to be insertion-ordered!

Hash Collision Resolution: Perturbation Probing

When two keys produce identical hash table indices, CPython resolves collisions via **open addressing** with a pseudo-random perturbation sequence:

$$\text{perturb} \gg= 5$$

$$j = (5 \times j + 1 + \text{perturb}) \pmod{\text{mask}}$$

This probe sequence ensures that all slots in the table are systematically visited without clustering.

Hashability Requirements

For an object to be used as a dictionary key or set element, it must satisfy the **Hash Contract**:

1. Implement `__hash__()` returning an integer that never changes during the object's lifetime.
2. Implement `__eq__()` such that if `a == b`, then `hash(a) == hash(b)`.
3. If an object is mutable, modifying it would alter its hash and break table lookups. Hence, mutable types (`list`, `dict`, `set`) unhashable: `TypeError: unhashable type: 'list'`.

4. Sets and Frozensets

A `set` in CPython is essentially a dictionary where the entries contain only keys and no values.

- **Lookup / Insert / Delete:** $O(1)$ average.
- **Set Operations:**
 - Union (`A | B`): $O(\text{len}(A) + \text{len}(B))$
 - Intersection (`A & B`): $O(\min(\text{len}(A), \text{len}(B)))$
 - Difference (`A - B`): $O(\text{len}(A))$
 - Symmetric Difference (`A ^ B`): $O(\text{len}(A) + \text{len}(B))$

5. High-Performance Alternatives: `collections` & `heapq`

`collections.deque` (Double-Ended Queue)

When you need to push and pop from both sides, never use `list.insert(0, x)` or `list.pop(0)`. Use `collections.deque`.

- In CPython, `deque` is implemented as a **doubly-linked list of fixed-size blocks** (each block holds 64 elements).
- Adding or removing from either head or tail is strictly $O(1)$ **without resizing copies**.

heapq : Priority Queues

heapq implements a **min-heap** directly on top of standard Python lists using 0-based array indexing:

- For node at index i :
 - Left child: $2i + 1$
 - Right child: $2i + 2$
 - Parent: $\lfloor (i - 1) / 2 \rfloor$
- Operations:
 - `heapify(lst)` : Transforms list into heap in-place in $O(N)$ time (sift-down algorithm).
 - `heappush(heap, item)` : $O(\log N)$.
 - `heappop(heap)` : $O(\log N)$.

6. Comprehensive Big-O Reference Matrix

Data Structure	Access by Index	Search by Value	Insert at Head	Insert at Tail	Delete at Head	Delete at Tail
<code>list</code>	$O(1)$	$O(N)$	$O(N)$	$O(1)$ amortized	$O(N)$	$O(1)$
<code>deque</code>	$O(N)$	$O(N)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
<code>dict</code>	N/A	$O(1)$ key lookup	N/A	$O(1)$ amortized	N/A	$O(1)$
<code>set</code>	N/A	$O(1)$ membership	N/A	$O(1)$ insert	N/A	$O(1)$ pop
<code>heapq</code>	$O(1)$ (min item)	$O(N)$	N/A	$O(\log N)$ push	$O(\log N)$ pop	N/A

Practice Problems & Interview Drills

Drill 1: The `[[[]] * 5]` Multiplication Trap

What bug does this code introduce, and how does pointer layout explain it?

Explanation & Solution: The outer `* 3` does not clone the inner list; it copies the **pointer** to that same single list three times!

Drill 2: Implement an LRU Cache from Scratch

Requirement: Implement a Least Recently Used (LRU) Cache with $O(1)$ `get()` and $O(1)$ `put()`. Do not use `collections.OrderedDict`.

Architecture: Combine a **hash map** (for $O(1)$ key-to-node lookup) with a **doubly linked list** (for $O(1)$ node removal and insertion to the front).

Version Notes

- Dict insertion order is a language guarantee from 3.7; the `|` merge operator for dicts needs 3.9.
- `itertools.batched` (3.12) replaces hand-written chunking helpers like the one in the Chapter 01 exercise; `functools.cache` needs 3.9.

Exercises

Three graded exercises for this chapter (two coding, one debugging) with hidden tests:

Attempt first; the reference solutions are in `exercises/_answers/ch02.py`.

Further Reading

- [Data structures tutorial](#)
- [collections module](#)
- [TimeComplexity wiki](#)

Check Yourself

Answer in your head or on paper first, then open each answer.

▼ 1. What is the average time complexity of `x` in `some_set` versus `x` in `some_list`?

`O(1)` average for a set (hash lookup) versus `O(n)` for a list (linear scan).

▼ 2. Why must dict keys be hashable, and what rule links `__hash__` and `__eq__`?

The key's hash picks a slot. Objects that compare equal must have equal hashes (`a == b` implies `hash(a) == hash(b)`), otherwise lookups fail.

▼ 3. Why is `list.insert(0, x)` slow but `collections.deque.appendleft(x)` fast?

A list is a contiguous array, so inserting at the front shifts all elements (`O(n)`). A deque is a linked structure of blocks with `O(1)` operations at both ends.

▼ 4. Since which version do dicts preserve insertion order as a language guarantee?

Python 3.7 (it was an implementation detail of CPython 3.6).